

UR3e

Collaborative Robot

QUICK-START GUIDE

Startup · Basic Operation · Shutdown · Troubleshooting

For New Operators

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Introduction

About the UR3e

The UR3e is a compact, six-axis collaborative robot designed to perform industrial tasks. Its robotic arm can be integrated into a variety of automation applications, including applications in which people and robots may share a workspace when appropriate safety measures are implemented.

Purpose of This Guide

This guide provides operators with the fundamental information needed to use the UR3e. It describes the tasks required to prepare, start, operate, and shut down the system and provides guidance for addressing basic operational issues.

Intended Audience

This guide is intended for trained operators who are new to the UR3e. Operators should already understand basic workplace safety, established safety procedures, and the work environment in which the robot will operate.

Safety

General Safety Guidelines

Before Operating the UR3e:

- Follow the established safety procedures for the specific robot application.
- Check that the work area is clear of hazards before operating the robot.
- Identify the location of the emergency stop before operating the robot.

- Do not operate the system if you identify damage or an unsafe condition.
- Do not change safety settings unless authorized.
- Stop operation if the UR3e behaves unexpectedly. Follow established safety procedures before attempting to identify or correct the issue.

Emergency Stop

The Emergency Stop is a red push-button on the Teach Pendant. Press the Emergency Stop to stop robot motion during an emergency. Do not use the Emergency Stop as a substitute for normal stopping procedures. After activation, the button remains engaged until it is manually reset.

Safety Symbols and Conventions

Signal Word	Meaning	Example
WARNING	Indicates a potentially hazardous situation that could result in serious injury or death.	An improperly secured robot can become unstable during operation, resulting in serious injury.
CAUTION	Indicates a potentially hazardous situation that could result in minor or moderate injury or equipment damage.	Do not extend the robot connection by joining Robot Cables. An improper connection may cause loss of power to the robot arm.
NOTE	Indicates important information that helps the user complete a task or understand the system.	Use the PolyScope interface on the Teach Pendant to access robot controls and programming functions.

UR3e System Overview

System Components

The UR3e system consists of several components that work together to control and operate the robot.

Component	Function
Robot Arm	Performs precise, automated movements for tasks such as assembly, screwdriving, polishing, and dispensing.
Control Box	Controls the robot's operation and supplies power to the robot arm.
Teach Pendant	Allows the operator to control, program, and monitor the robot through the PolyScope interface.
Robot Cable	Connects the robot arm to the Control Box and carries power and communication signals between the components.

Robot Arm

The UR3e robot arm has six rotating joints that allow the arm to position and orient its tool in different ways. The UR3e has a maximum payload of 3 kg and a reach of 500 mm. With a pose repeatability of ± 0.03 mm, the robot can return to programmed positions with a high degree of consistency. The primary components of the UR3e arm include the base, six rotating joints and tool flange (see Figure 1).

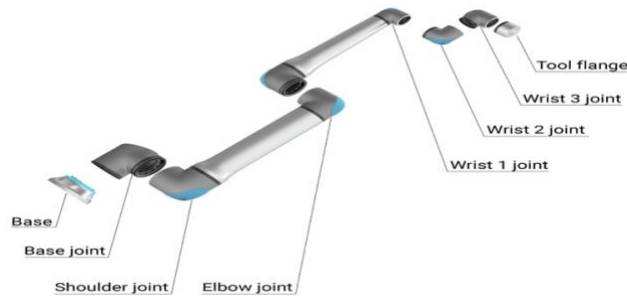


Figure 1. UR3e robot arm. Source: Universal Robots

Teach Pendant and PolyScope

The Teach Pendant is a handheld touchscreen device used to interact with the UR3e. PolyScope is the graphical interface used to control and program the robot. Operators can use PolyScope to monitor operation, modify settings, and view the robot's status.

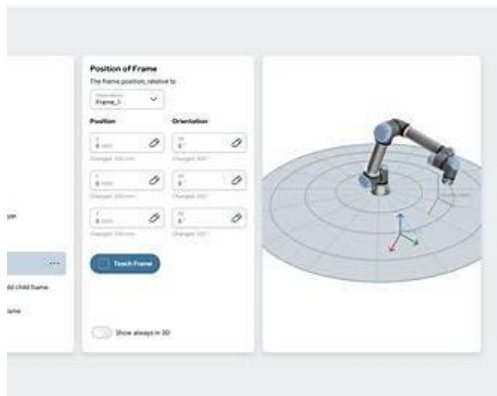


Figure 2. PolyScope graphical user interface. Source: Universal Robots

Before You Begin

Pre-Operation Check

Before operating the UR3e, complete the following checks to verify that the system is ready for operation.

- ☐ Inspect the robot arm for visible damage.
- ☐ Verify that all system components are securely connected.
- ☐ Check that the work area is clear of obstacles and hazards.
- ☐ Verify that all required safety functions are active.
- ☐ Confirm that the payload is correctly configured for the application.

Work Area

Before operating the UR3e, inspect the surrounding work area to ensure the robot can move without obstruction.

- ☐ Remove objects that could interfere with the robot's movement.
- ☐ Keep unauthorized personnel outside the designated work area.
- ☐ Verify that required safeguards are correctly positioned and active.
- ☐ Check for environmental conditions that could interfere with safe operation.

System Connections

Before powering on the UR3e, verify that all required system connections are secure.

- ☐ Confirm that the Robot Cable is securely connected to the robot arm and Control Box.
- ☐ Inspect cables and connectors for visible damage.
- ☐ Verify that required power and system connections are secure.
- ☐ Do not operate the robot with damaged or improperly connected cables.

Powering On the System

Power-On Procedure

Complete the pre-operation checks in Section 4 before powering on the UR3e.

1. Verify that the Control Box is connected to an appropriate power source.
2. Confirm that the Robot Cable is securely connected to the robot arm and Control Box.
3. Press the power button on the Teach Pendant.

4. Allow the system to start and wait for the PolyScope interface to appear.
5. Check the system status for warnings, faults, or other conditions requiring attention.

NOTE: Powering on the system does not automatically enable robot movement. The robot arm must be initialized before operation.

Power-On Verification

Before proceeding, verify that the Teach Pendant and PolyScope interface are operating normally. Address any displayed warnings or faults according to approved procedures before initializing the robot arm.

Initializing the Robot Arm

Before Initialization

Before initializing the robot arm, ensure that the active payload and installation settings are accurate. Make sure the robot arm is not touching nearby objects and move outside the robot's workspace before releasing the brakes.

WARNING: The robot arm may make slight movements as the joint brakes are released. Keep clear of the robot's workspace during initialization.

Initialization Procedure

1. Find the Robot State icon in the PolyScope interface.
2. Open the Initialize window and select **Power on**.
3. Ensure that the active payload is correct for the current application.
4. Select **Unlock** to release the robot's joint brakes.
5. Wait for the initialization process to finish.
6. Verify that the robot state displays **Active** before you proceed.

Basic Operation

Operating the UR3e

Before starting robot movement, verify that the UR3e is initialized and the correct program is selected for the application.

1. Confirm that the robot status is ready for operation.
2. Select the approved program for the current task.
3. Verify that the work area is clear.
4. Start the program using the PolyScope controls.
5. Observe the robot as it begins the programmed operation.

WARNING: Do not enter the robot's operating area while the robot is moving unless the application has been specifically assessed and configured for that interaction.

Monitoring Operation

Monitor the UR3e throughout operation. Observe the robot's movement and check the PolyScope interface for warnings, faults, or unexpected changes in system status.

Stop operation if the robot behaves unexpectedly or if an unsafe condition develops.

Stopping Operation

Use the normal stop controls in PolyScope to stop the robot during routine operation. Allow the robot to stop before entering its operating area or performing another task.

Use the Emergency Stop only when an emergency requires robot motion to stop immediately.

Shutting Down the System

Normal Shutdown

Use the normal shutdown procedure when operation is complete.

1. Stop the active robot program and verify that robot movement has stopped.
2. Tap the **Robot State** icon in PolyScope to turn off the robot arm.
3. Verify that the Robot State icon changes from green to white.
4. Press the power button on the Teach Pendant to turn off the Control Box.
5. If the **Shutdown** dialog appears, select **Power Off**.

CAUTION: Do not use the Emergency Stop as a substitute for the normal shutdown procedure.

After Shutdown

After the system powers down, verify that the robot arm and Control Box are off. If the system must be disconnected from its power source, allow at least 30 seconds for stored energy to discharge before proceeding.

Troubleshooting

Basic Troubleshooting

Use the following table to identify and address basic operational issues. Stop operation and follow established safety procedures if an issue cannot be resolved safely.

Issue	Possible Cause	Recommended Action
Robot arm does not initialize	Incorrect payload or installation settings	Incorrect payload or installation settings
Robot state remains Locked	Joint brakes have not been released	Select Unlock in the Initialize window and wait for the robot state to display Active .
Protective Stop occurs	Robot has approached or exceeded an operating or safety limit	Review the warning displayed in PolyScope and investigate the cause before resuming operation.
Robot behaves unexpectedly	Incorrect configuration, program, or application condition	Stop operation. Check the system status and follow the approved troubleshooting procedure before restarting.

Technical Specifications

UR3e Specifications

Specification	Value
Maximum payload	3 kg (6.6 lb)
Reach	500 mm (19.7 in)
Degrees of freedom	6 rotating joints
Pose repeatability	±0.03 mm
Robot arm weight	11.2 kg (24.7 lb)
Maximum TCP speed	3 m/s
IP classification	Ip54
Tool I/O power supply	12 V/24 V, 600 mA
Operating Temperature	Operating temperature → 0–50 °C (32–122 °F)

References

The following Universal Robots resources were consulted during the development of this guide:

1. **Universal Robots. UR3e User Manual.**
Manufacturer documentation for UR3e installation, operation, safety, initialization, and system specifications.
2. **Universal Robots. UR3e Product Specifications.**
Manufacturer specifications for payload, reach, repeatability, dimensions, operating conditions, and other technical characteristics.
3. **Universal Robots. PolyScope X User Manual.**
Manufacturer documentation for the PolyScope interface, robot initialization, system status, operation, and shutdown procedures.
4. **Universal Robots. Safety Information and Technical Documentation.**
Manufacturer guidance concerning safe operation, emergency stopping, protective stops, and safety functions.

Manufacturer documentation: Universal Robots